| | |
|---|---|
| **Student Name** | farjana .s |
| **Register Number** | TU-6253210211012 |
| **Project Title** | **AI-Based Employee Attrition Prediction System** |
| **Week** | Week 2 |

# WEEK-2 TASK

## Task Name

**AI-Based Employee Attrition Prediction System**

## Abstract

An AI-Based Employee Attrition Prediction System uses machine learning algorithms to predict whether an employee is likely to leave an organization. The system analyzes employee information such as age, salary, years of experience, job satisfaction, overtime, and performance ratings. This helps companies identify employees at risk of leaving and take preventive measures to improve employee retention.

---

## Introduction

Employee attrition is a major challenge for organizations because replacing skilled employees is expensive and time-consuming. By using Artificial Intelligence and Machine Learning, organizations can predict employee turnover in advance. The system learns from historical employee data and provides accurate predictions that help HR departments make informed decisions.

---

## Objectives

1. To predict employee attrition using machine learning.
2. To improve employee retention strategies.
3. To reduce hiring and training costs.

4. To identify factors affecting employee turnover.
5. To assist HR departments in decision-making.

---

# Python Code

```
from sklearn.model_selection import train_test_split

from sklearn.tree import DecisionTreeClassifier

from sklearn.metrics import accuracy_score


# Sample Dataset

X = [

    [25, 30000, 1],

    [40, 70000, 0],

    [30, 50000, 1],

    [35, 60000, 0],

    [28, 35000, 1],

    [45, 80000, 0]

]


# 1 = Leave, 0 = Stay

y = [1, 0, 1, 0, 1, 0]


X_train, X_test, y_train, y_test = train_test_split(

    X, y, test_size=0.3, random_state=42

)


model = DecisionTreeClassifier()
```

```
model.fit(X_train, y_train)


prediction = model.predict(X_test)


print("Prediction:", prediction)

print("Accuracy:", accuracy_score(y_test, prediction))
```

---

# Simulation Results & Observations

### Prediction Efficiency

The model successfully predicts whether an employee is likely to leave based on employee information.

### Accuracy

The Decision Tree algorithm provides reliable predictions using historical employee data.

### Benefits

- Early identification of employees at risk.
- Improved employee retention.
- Reduced recruitment costs.
- Better workforce planning.
- Data-driven HR decisions.

---

# Conclusion

The AI-Based Employee Attrition Prediction System demonstrates how machine learning can support Human Resource Management. By predicting employee turnover, organizations can improve employee satisfaction, reduce attrition, and make better workforce management decisions.